Table A1: Detailed Ablation Study on OOD detection.

$\mathcal{L}_n$	$\mathcal{L}_g$	$\mathcal{L}_{sp}$	$\mathcal{L}_{cp}$	LowR	HHM	BZR	PTC-MR	AIDS	ENZYMES	IMDB-M	Tox21	FreeSolv	BBBP	ClinTox	Esol
						COX2	MUTAG	DHFR	PROTEIN	IMDB-B	SIDER	ToxCast	BACE	LIPO	MUV
✓	-	-	-	-	-	83.51±4.14	72.48±3.77	96.84±0.58	60.85±2.95	79.34±1.81	62.58±0.67	59.48±2.20	69.53±2.29	53.29±4.32	86.49±1.20
-	✓	-	-	-	-	87.44±4.66	77.84±3.71	97.60±1.05	56.74±1.96	75.22±1.91	65.07±1.32	78.40±6.44	77.66±2.29	70.11±2.44	89.57±2.80
-	-	✓	-	-	-	79.21±5.60	74.83±8.54	89.47±1.85	50.43±7.41	72.91±2.75	54.84±2.56	58.16±6.23	58.09±5.43	58.46±5.35	83.35±2.71
-	-	-	✓	-	-	28.70±1.28	84.86±1.35	78.47±2.65	46.18±8.40	69.93±7.49	50.39±31.2	43.08±1.43	44.67±2.15	56.57±5.40	71.65±1.13
✓	✓	-	-	-	-	93.14±3.63	77.53±4.02	98.90±0.42	61.48±3.46	79.55±1.35	65.44±1.13	71.45±4.23	80.43±2.57	65.89±4.57	90.94±1.16
✓	-	✓	-	-	-	85.01±3.05	76.10±3.01	96.87±0.52	59.69±1.89	79.69±1.67	63.01±0.97	56.30±5.33	69.66±2.45	54.14±4.01	86.31±1.99
✓	-	-	✓	-	-	84.44±4.55	79.96±3.55	90.16±14.43	49.54±5.35	81.29±0.24	61.37±3.43	51.05±3.58	57.85±3.81	51.12±3.51	80.64±1.25
-	✓	✓	-	-	-	86.59±5.24	77.97±4.00	97.22±1.35	55.51±4.39	76.17±1.65	65.48±0.78	77.38±5.19	79.77±4.39	70.20±1.01	88.33±1.35
-	✓	-	✓	-	-	89.77±37.10	84.24±0.82	89.86±16.10	53.37±9.46	78.20±2.30	65.52±3.41	59.54±2.90	58.33±4.61	65.44±1.21	79.27±2.16
-	-	✓	✓	-	-	78.28±6.06	80.00±3.84	87.46±10.40	52.68±8.96	68.88±4.54	65.22±1.93	51.06±10.13	74.48±5.71	61.53±1.89	81.15±7.33
✓	✓	✓	-	✓	✓	94.99±2.25	81.21±2.65	99.07±0.40	61.84±1.94	79.94±1.09	66.50±1.35	80.13±3.43	82.91±2.58	69.18±3.61	91.52±0.70
✓	✓	-	✓	✓	✓	93.11±1.06	90.00±0.45	96.30±0.73	63.21±7.93	73.11±9.32	36.05±3.51	45.80±5.61	81.04±1.37	74.02±1.07	75.19±0.45
-	✓	✓	✓	✓	✓	95.46±2.08	89.47±2.31	95.20±0.53	60.91±3.53	79.76±1.41	63.17±1.43	62.52±2.03	77.30±1.47	74.44±0.00	83.22±1.22
✓	✓	✓	✓	✓	✓	**98.00±0.62**	**91.96±3.14**	**99.38±0.09**	**67.38±2.30**	**81.90±0.98**	**68.48±0.10**	**80.78±0.30**	**84.27±1.60**	**75.87±1.25**	**92.93±0.77**

A Additional Experimental Analysis

In this section, we provide more detailed experiments to further validate the effectiveness of our proposed components.

A.1 Ablation Study

To analyze the individual contributions of different components to the overall performance gain, we conducted a comprehensive suite of ablation studies derived from the full-featured framework.

It can be observed that removing the total loss of Cross-branch prototype contrast (w/o $\mathcal{L}_p$) leads to a dramatic performance decline on several datasets (e.g., BZR and COX2), indicating that $\mathcal{L}_p$ is crucial for capturing discriminative patterns for anomaly detection. Similarly, the absence of the graph-level loss (w/o $\mathcal{L}_g$) or the sample-prototype contrastive loss (w/o $\mathcal{L}_{sp}$) results in a consistent drop in accuracy, which suggests that these losses provide essential supervision for learning robust graph representations.

Furthermore, we investigate the impact of the structural modules. Excluding the Low-Rank constraint (w/o LowR) causes a noticeable degradation in performance and, in some cases, increases the standard deviation. This indicates that LowR plays a vital role in regularizing the feature space and improving model stability. Notably, removing the Hypergraph Homophily Module (w/o HHM) also yields inferior results compared to the full model, demonstrating that HHM effectively exploits high-order structural information to distinguish OOD graphs. The full model achieves optimal performance across all datasets, confirming that hypergraph-based high-order modeling is inherently superior to traditional modeling for robust graph OOD detection.

A.2 Time Complexity Study

Thanks for your valuable feedback! We agree that the original manuscript did not sufficiently discuss computational complexity and practical efficiency. Specifically, the training complexity of HGOOD can be summarized as:

$$\mathcal{O}\big(TN_g[Ld_m(\tilde{m}+\tilde{n}) + IKd_m + \tilde{n}^2 + L'\tilde{n}d_m + cd_m]\big),$$

where the $\tilde{n}^2$ term associated with hypergraph construction is non-negligible for dense graphs. To further support this point, we additionally report the time and memory statistics.

No. of Nodes	100	500	1000	2500	4000	7000	10000
Time (s)	0.53	1.07	1.84	8.94	23.67	72.14	146.82
Memory (GB)	0.89	1.13	1.36	1.88	2.44	3.58	4.83

As the graph size increases, the time cost grows much faster than the memory cost, which is consistent with the quadratic overhead of hypergraph construction, while the memory growth remains relatively moderate due to our low-rank parameterization.

A.3 Robustness Study

To further validate the generalization ability of our method, we conduct additional robustness experiments on two real-world molecular datasets, namely BZR and COX2. Specifically, we manually inject **symmetric label noise** with different noise rates p to simulate annotation errors, and adopt random **node masking** with various node removal ratios to mimic incomplete graph structures. The experimental results illustrate that our proposed HGOOD model **consistently achieves the best performance under various kinds of data perturbations**, which clearly demonstrates its superior robustness and strong resistance to diverse disturbances in real-world applications.

Setting	$p/ratio$	GOODAT	CGOD	HGOOD (ours)
Clean	—	96.43	98.17	99.38
Symmetric Label Noise	10%	92.12	94.56	97.54
	20%	80.51	83.24	89.58
	40%	65.20	68.93	78.35
Node Masking	5%	91.53	93.81	96.39
	10%	77.42	79.85	85.37
	15%	57.90	60.32	73.61

B Dataset Description

B.1 Distribution Shift Types

We summarize the employed distribution shift types and the corresponding ID/OOD dataset pairs in the following table.

Shift Type	Included ID/OOD Dataset Pairs
Feature Shift	BZR/COX2, AIDS/DHFR, Tox21/SIDER
Structure Shift	PTC-MR/MUTAG, IMDB-M/IMDB-B
Mixed Shift	ENZYMES/PROTEIN

B.2 Statistics Of Datasets

We summarize the detailed statistics of all datasets in the following table, including the number of node features, the total number of graphs, as well as the average number of nodes, edges, and node degree.

Dataset	#Feature	#Graphs	Avg. Nodes	Avg. Edges	Avg. Deg.
AIDS	1	2000	15.69	16.20	1.03
DHFR	1	756	42.43	44.54	1.05
ENZYMES	1	600	32.63	62.13	1.90
PROTEIN	1	1113	39.06	72.82	1.86
IMDB-M	1	1500	13.00	65.93	5.07
IMDB-B	1	1000	19.77	96.53	4.88
Tox21	9	7831	18.57	19.29	1.04
SIDER	9	1427	33.64	35.35	1.05
FreeSolv	9	642	8.72	8.38	0.96
ToxCast	9	8576	18.78	19.26	1.03
BBBP	9	2039	24.06	25.95	1.08
BACE	9	1513	34.08	36.85	1.08
ClinTox	9	1477	26.15	27.88	1.07
LIPO	9	4200	27.04	29.49	1.09
Esol	9	1128	13.28	13.67	1.03
MUV	9	93087	24.23	26.27	1.08
BZR	1	405	35.75	38.36	1.07
COX2	1	467	41.22	43.45	1.05
PTC_MR	1	344	14.28	14.69	1.03
MUTAG	1	188	17.93	19.79	1.10

C Overall Optimization and OOD Scoring

Algorithm 1 The overall training process of HGOOD

Input: Training dataset $\mathcal{G}^{id}_{train}$; hyper-parameters c, K, α, τ.
Output: Parameters of GIN encoder $\mathcal{F}_s(\cdot)$ and $\mathcal{F}_b(\cdot)$, hypergraph projection matrix $\boldsymbol{P}$, hypergraph weight matrix $\boldsymbol{W}$.

1: Generate feature and structure views (G_b, G_s) via perturbation-free graph data augmentation.
2: **while** not convergence **do**
3: Encode feature and structure information of the graph based on G_b and G_s to get $\boldsymbol{h}_i^{(b)}$,$\boldsymbol{h}_i^{(s)}$; `// Eq. 3`
4: Calculate the graph feature and structure embeddings $\boldsymbol{h}_G^{(b)}$,$\boldsymbol{h}_G^{(s)}$; `// Eq. 4`
5: Construct hypergraph structure dependency $\boldsymbol{\rho}$; `// Eq. 5`
6: Perform l hypergraph convolution to get hypergraph node embedding $\tilde{\boldsymbol{\Lambda}}_l$; `// Eq. 6`
7: Calculate the hypergraph embedding $\boldsymbol{h}_G^{\text{global}}$; `// Eq. 7`
8: Compute node-level and graph-level contrastive learning loss $\mathcal{L}_n, \mathcal{L}_g$; `// Eqs. 8-9`
9: Apply k-means clustering separately to graphs and hypergraphs, and derive prototypes C^{local} and C^{global};
10: Compute the sample-prototype contrastive loss and the cross-prototype contrastive loss ; `// Eqs. 10-13`
11: Calculate the standard deviations of the losses and adopt an adaptive weighted loss strategy ; `// Eqs. 14`
12: **end while**
13: Obtain parameters of GIN encoder $\mathcal{F}_s(\cdot)$ and $\mathcal{F}_b(\cdot)$, hypergraph projection matrix $\boldsymbol{P}$ and hypergraph weight matrix $\boldsymbol{W}$.

In this section, we present the formal procedure for the training process of HGOOD.